Fakultas Informatika
School of Computing
Telkom University

Aplikasi Berbasis Platform

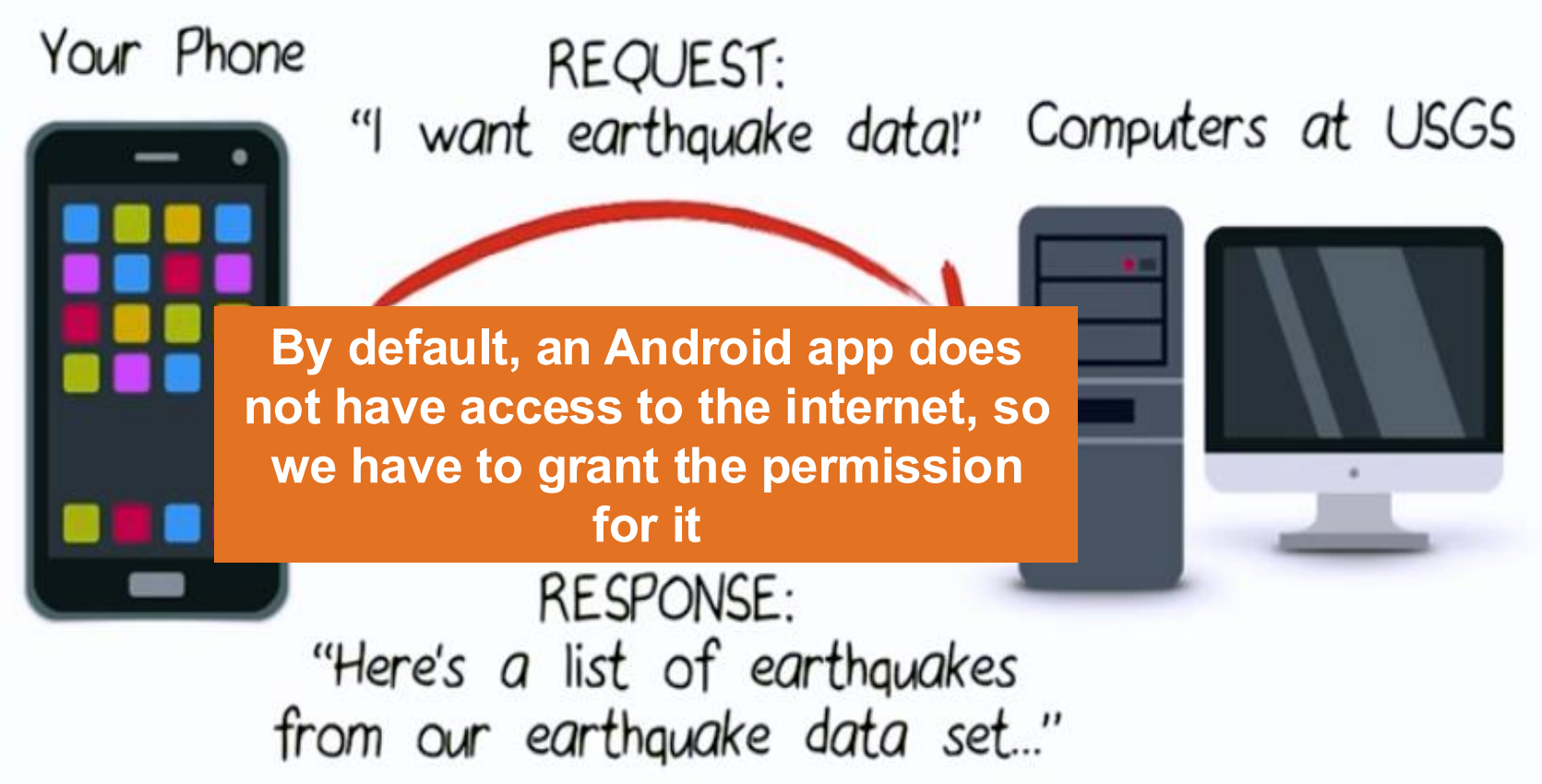
Networking

Outline

- Permissions in Android
- Review: Sync & Async Process
- Networking in Flutter

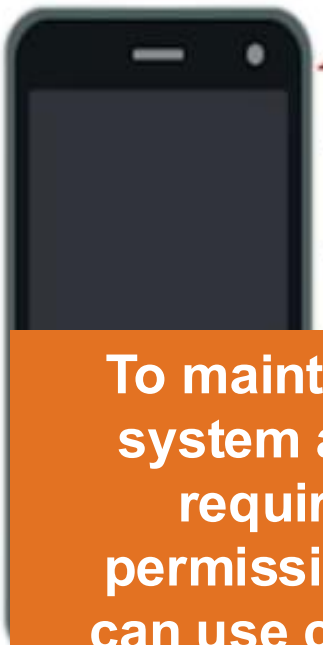
Permissions in Android

HTTP Request & Response



By default, an Android app does not have access to the internet, so we have to grant the permission for it

Permissions in Android



Use internet

In AndroidManifest.xml:

```
<uses-permission android:name="android.permission.INTERNET"/>
```

Send SMS message

Call phone number

Record Microphone Audio

Get current location

Network connectivity status

...and more!

To maintain security for the system and users, Android requires apps to request permission before the apps can use certain system data and features.

Level Of Permissions

Normal Permissions

Access the Internet
Vibrate the device
Set the timezone
Network connectivity status

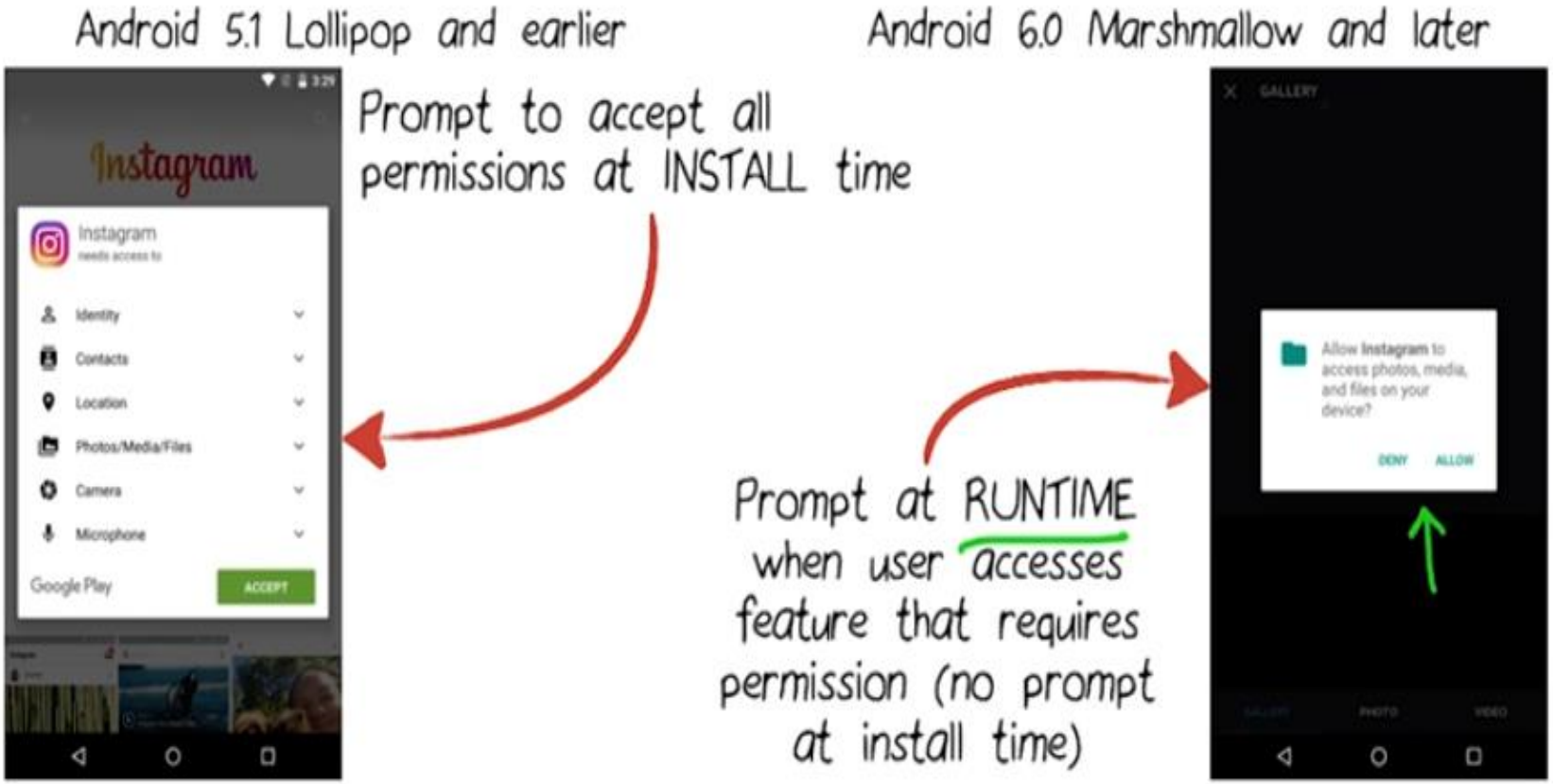
Automatically granted by the system

Dangerous Permissions

Use Camera
Access call log
Access Contacts
Record Audio

Requested at runtime when app needs the permission pop up a dialog to ask for permission

Backwards Compatibility



Review: Sync & Async Process

Review: Sync Process

- In computing, we say something is synchronous when it waits for an event to happen before continuing.
- A disadvantage in this approach is that if a part of the code takes too long to execute, the subsequent blocks, though unrelated, will be blocked from executing.

Review: Sync Process

- A synchronous execution model will block every other user's request till it finishes processing the current request. In such a case, like that of a web server, every request must be independent of the others.
- This means, the webserver should not wait for the current request to finish executing before it responds to request from other users.
- Simply put, it should accept requests from new users before necessarily completing the requests of previous users. This is termed as asynchronous.

Review: Async Process

- Asynchronous programming basically means no waiting or non-blocking programming model
- An asynchronous operation executes in a thread, separate from the main application thread
- When an application calls a method to perform an operation asynchronously, the application can continue executing while the asynchronous method performs its task

Main (Sync) vs Background (Async) Threads

Async Process Example

Networking in Flutter

Package

- Flutter supports using shared packages contributed by other developers to the Flutter and Dart ecosystems. This allows quickly building an app without having to develop everything from scratch.

Using Package

- Find Packages at pub.dev
- Open pubspec.yaml, add under “dependencies”
- Install it
  - Terminal: Run *flutter pub get* to show installed package or *flutter pub add package_name* to add new package
  - From android studio, when open pubspec click “pub get”
  - From VS Code: Click **Get Packages** located in right side of the action ribbon at the top of pubspec.yaml.
- Stop and Restart App, if necessary

Example

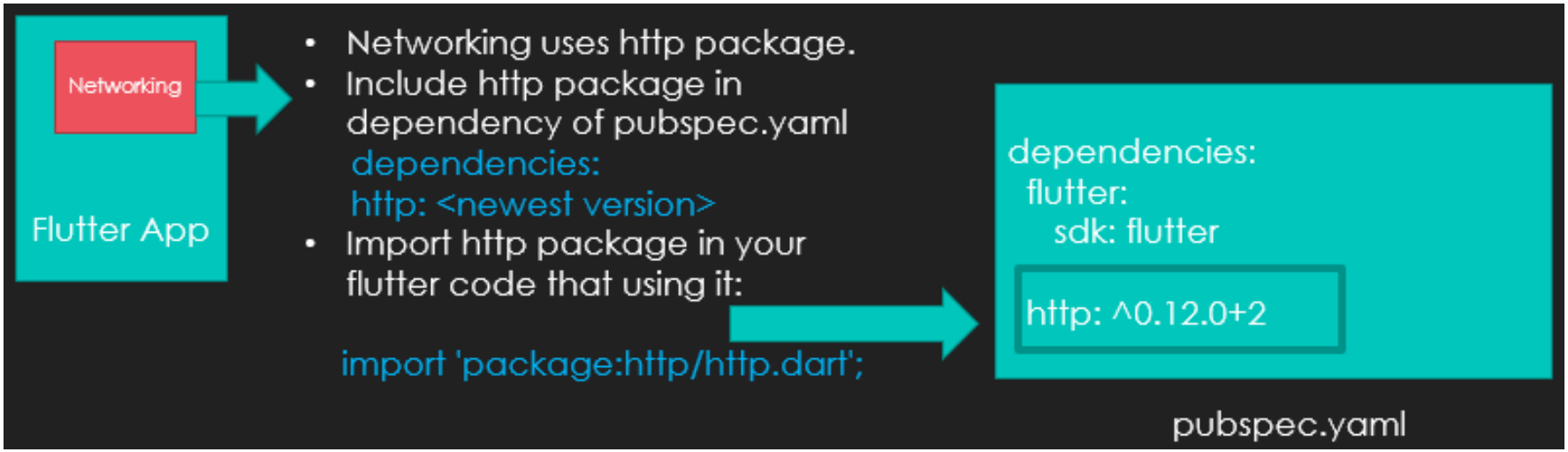
README.md | main.dart | pubspec.yaml

Flutter commands | Pub get | Pu

```yaml
  # The following adds the Cupertino Icons font to your application.
  # Use with the CupertinoIcons class for iOS style icons.
  cupertino_icons: ^1.0.2
  json_annotation: ^4.4.0

dev_dependencies:
  flutter_test:
    sdk: flutter
```

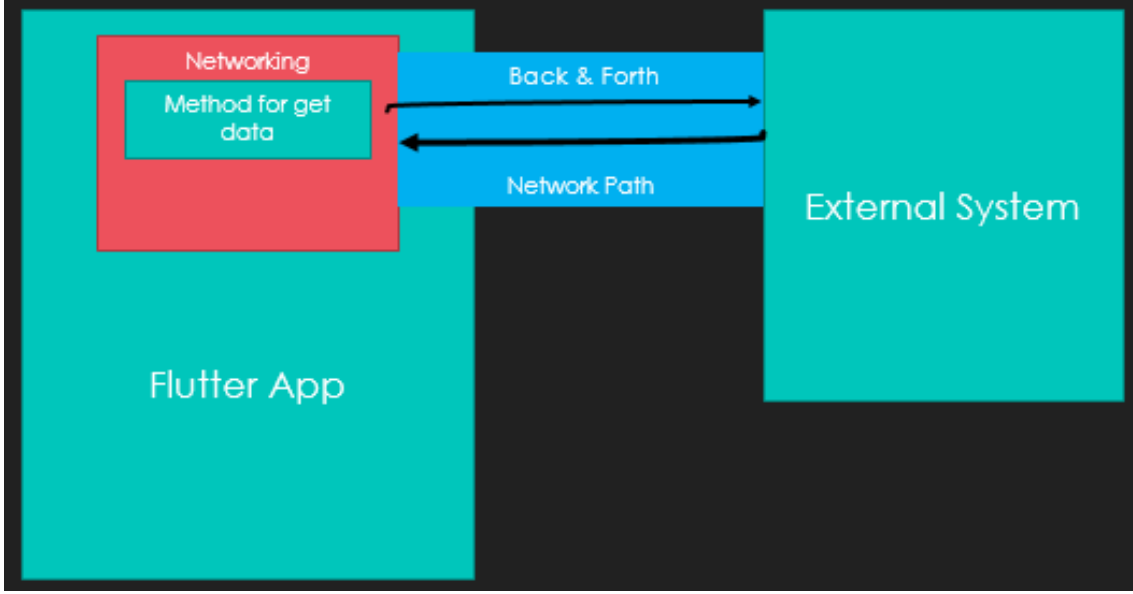
```dart
import 'package:flutter/material.dart';
import 'package:json_annotation/json_annotation.dart';

//import 'package:flutter/widgets.dart';
void main() {
  runApp(
    MaterialApp(home: AlertDial()), // use MaterialApp
  );
```

Add Networking Dependencies



Making Asynchronous Method

1. To get data from external system, Flutter needs a method that runs back and forth from app to external system

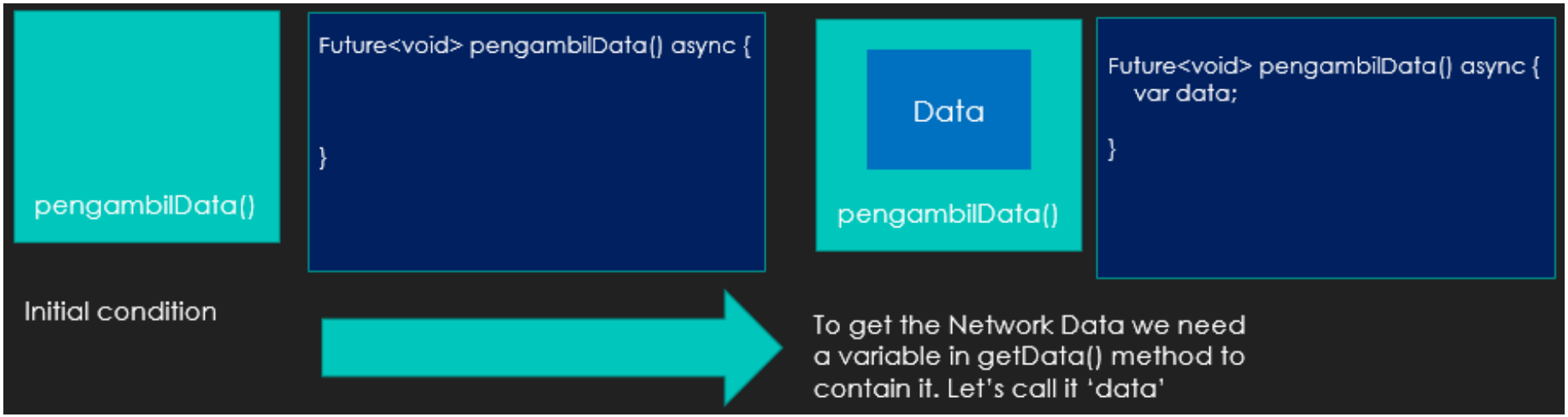


2. In order for the get data method to work asynchronously (parallel) and doesn't block main program, we need to make it with Future type. Future type method allows method to run in the background (asynchronous)

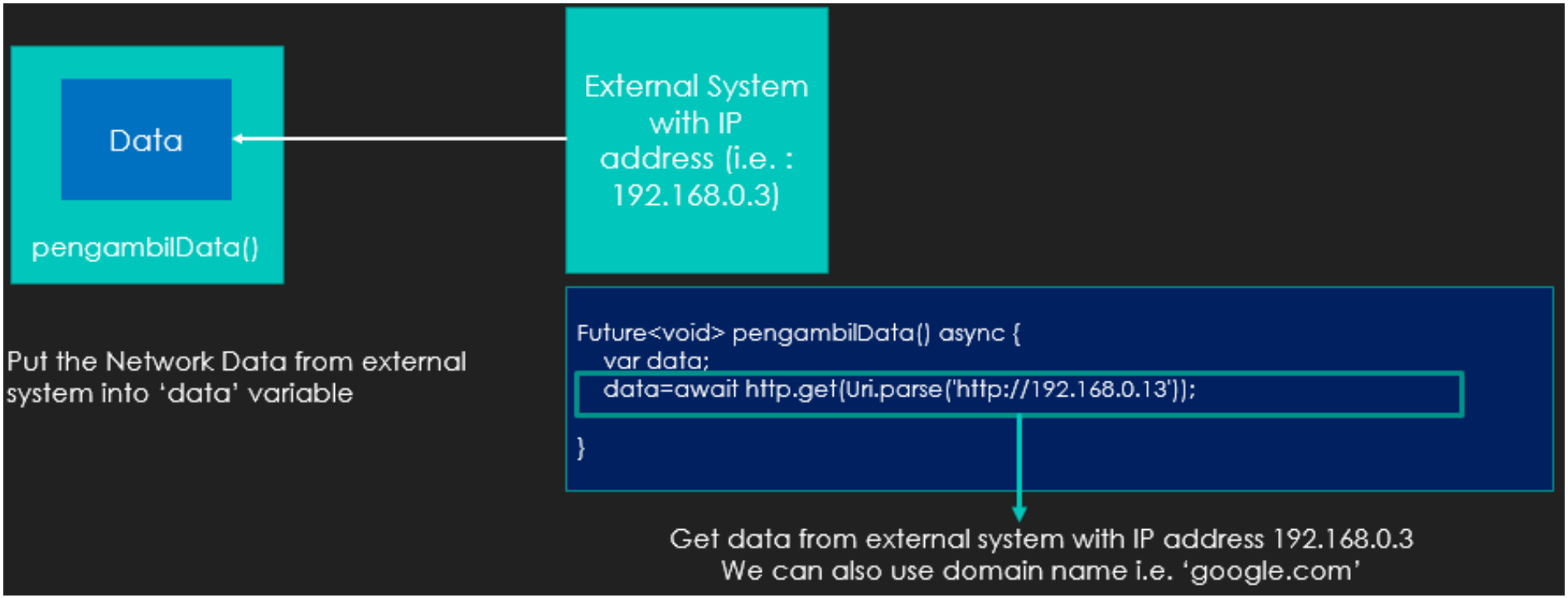
Future Type | Return Type | Method Name | Set to run Asynchronously

```
Future<void> pengambilData() async {
  //tempat kode
}
```

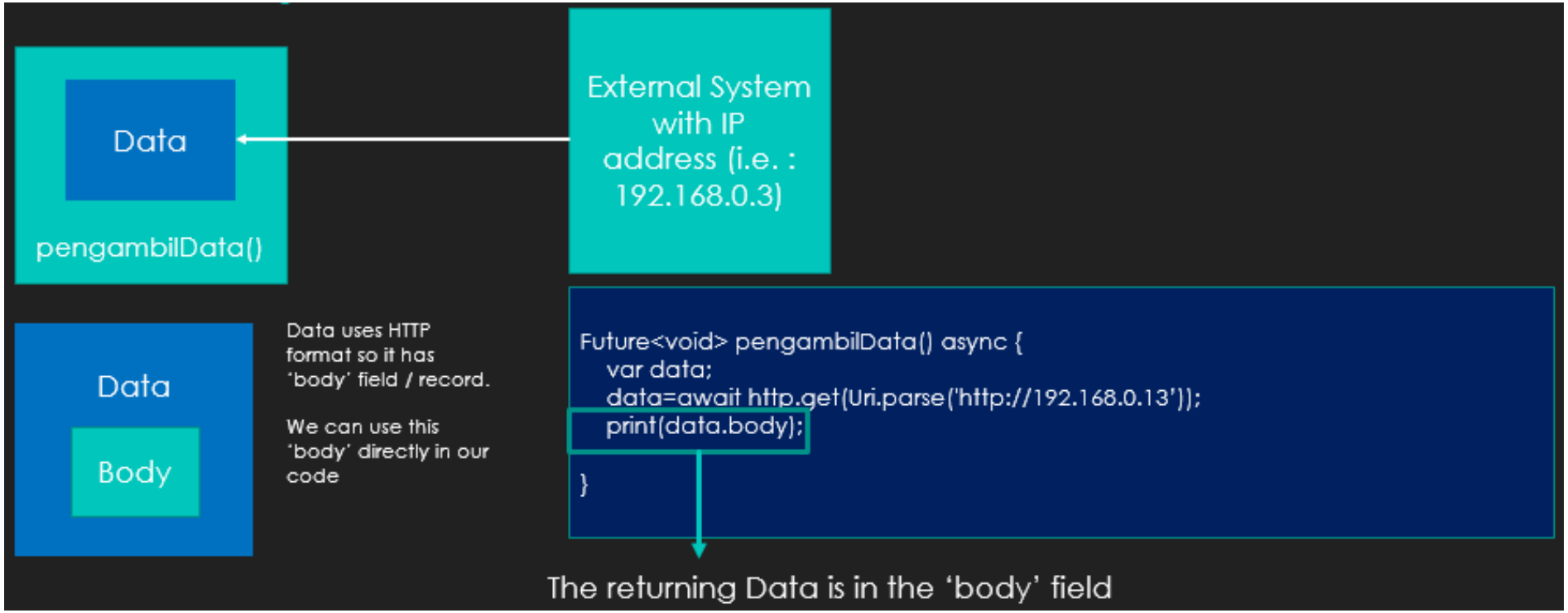
Get Data from External System



Get Data from External System



Get Data from External System



Dart Code

```dart
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() {
    pengambilData();
}

Future<void> pengambilData() async {
    var data;
    data = await http.get(Uri.parse('http://192.168.0.13/'));
    print(data.body);
}
```

Console

```
Performing hot restart...
Syncing files to device Android SDK built for arm64...
Restarted application in 853ms.
I/flutter ( 5375): tes dari html
```

Parsing JSON in Flutter

Let’s use some external system for test case: https://jsonplaceholder.typicode.com/posts


```
[
  {
    "userId": 1,
    "id": 1,
    "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
    "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
  },
  {
    "userId": 1,
    "id": 2,
    "title": "qui est esse",
    "body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non
debitis possimus qui neque nisi nulla"
  },
  {
    "userId": 1,
    "id": 3,
    "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
    "body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut"
  },
  {
    "userId": 1,
    "id": 4,
    "title": "eum et est occaecati",
    "body": "ullam et saepe reiciendis voluptatem adipisci\nsit amet autem assumenda provident rerum culpa\nquis hic commodi nesciunt rem tenetur doloremque ipsam iure\nquis sunt voluptate
rerum illo velit"
  },
  {
    "userId": 1,
    "id": 5,
    "title": "nesciunt quas odio",
    "body": "repudiandae veniam quaerat sunt sed\nalias aut fugiat sit autem sed est\nvoluptatem omnis possimus esse voluptatibus quis\nest aut tenetur dolor neque"
  },
  {
```

Parsing JSON in Flutter

- JSON structure:
  - userID
  - id
  - title
  - body


```
[
    {
        "userId": 1,
        "id": 1,
        "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
        "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas
            totam\nnostrum rerum est autem sunt rem eveniet architecto"
    },
```

Parsing JSON in Flutter

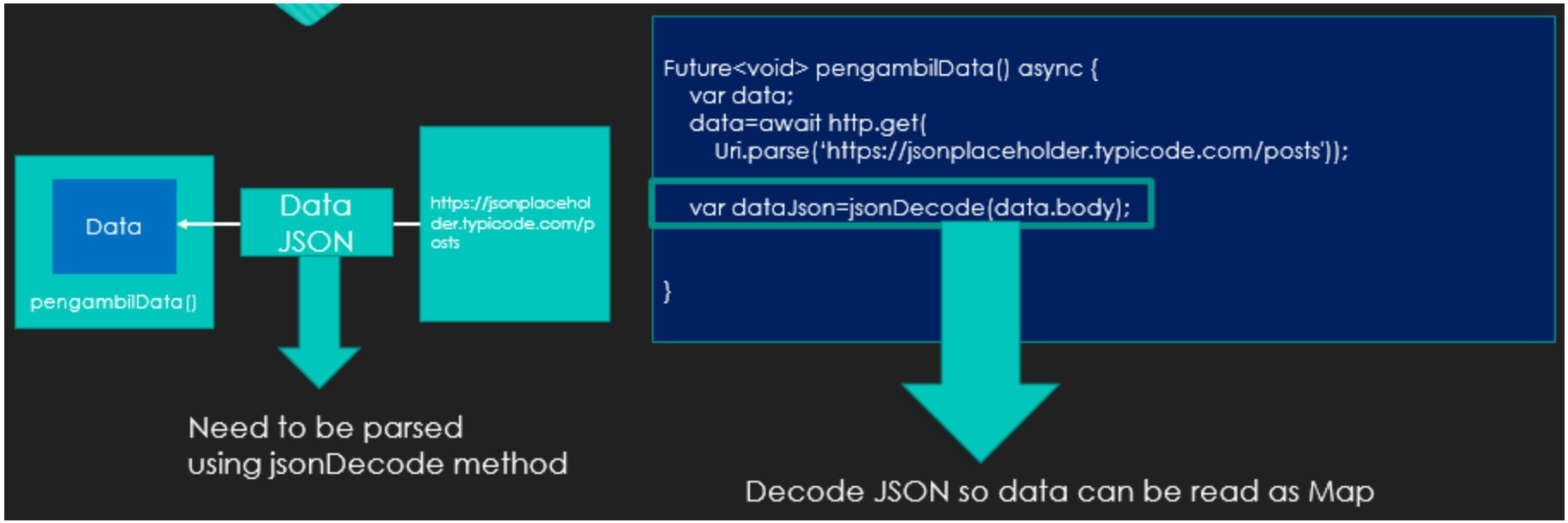
```dart
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() {
  pengambilData();
}

Future<void> pengambilData() async {
  var data;
  data = await http.get(Uri.parse('https://jsonplaceholder.typicode.com/posts'));
  print(data.body);
}
```

```
I/flutter ( 6250): [
I/flutter ( 6250):   {
I/flutter ( 6250):     "userId": 1,
I/flutter ( 6250):     "id": 1,
I/flutter ( 6250):     "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
I/flutter ( 6250):     "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet
 architecto"
I/flutter ( 6250):   },
I/flutter ( 6250):   {
I/flutter ( 6250):     "userId": 1,
I/flutter ( 6250):     "id": 2,
```

Using the previous code

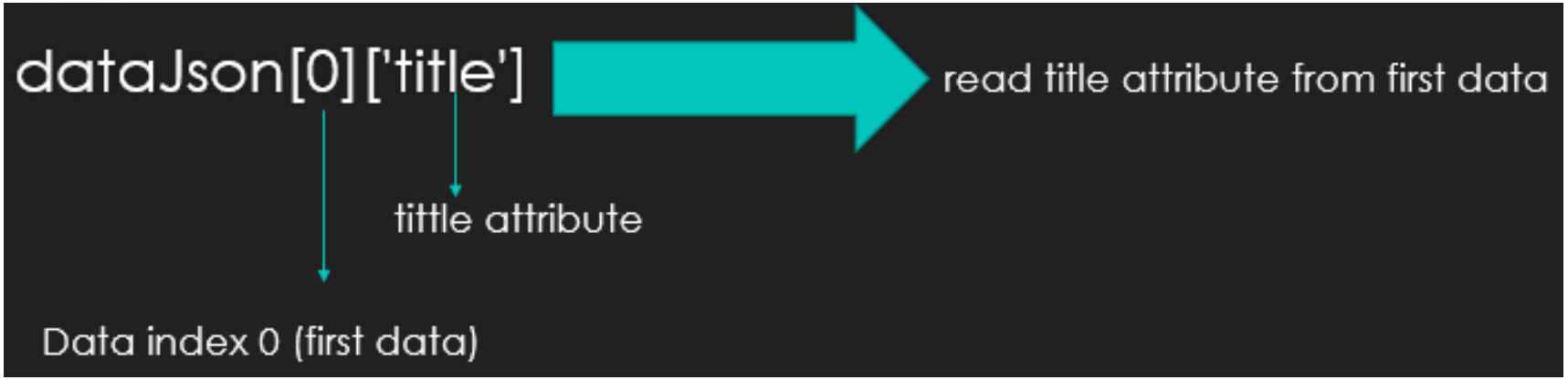
Data is still pure JSON

Parsing JSON in Flutter



Parsing JSON in Flutter

- In this case, the returning JSON data will be parsed as List of Map, so we can access it like this: NamaVariable[indexdata][namafield]



Parsing JSON in Flutter

```
Future<void> pengambilData() async {
   var data;
   data=await http.get(
      Uri.parse('https://jsonplaceholder.typicode.com/posts'));

   var dataJson=jsonDecode(data.body);
   print(dataJson[0]['userID']);
   print(dataJson[0]['id']);
   print(dataJson[0]['title']);
   print(dataJson[0]['body']);


}
```

Read all attributes from First Data

```
Performing hot restart...
Syncing files to device Android SDK built for arm64...
Restarted application in 478ms.
I/flutter ( 6250): null
I/flutter ( 6250): 1
I/flutter ( 6250): sunt aut facere repellat provident occaecati excepturi optio reprehenderit
I/flutter ( 6250): quia et suscipit
I/flutter ( 6250): suscipit recusandae consequuntur expedita et cum
I/flutter ( 6250): reprehenderit molestiae ut ut quas totam
I/flutter ( 6250): nostrum rerum est autem sunt rem eveniet architecto
```

Parsing JSON in Flutter

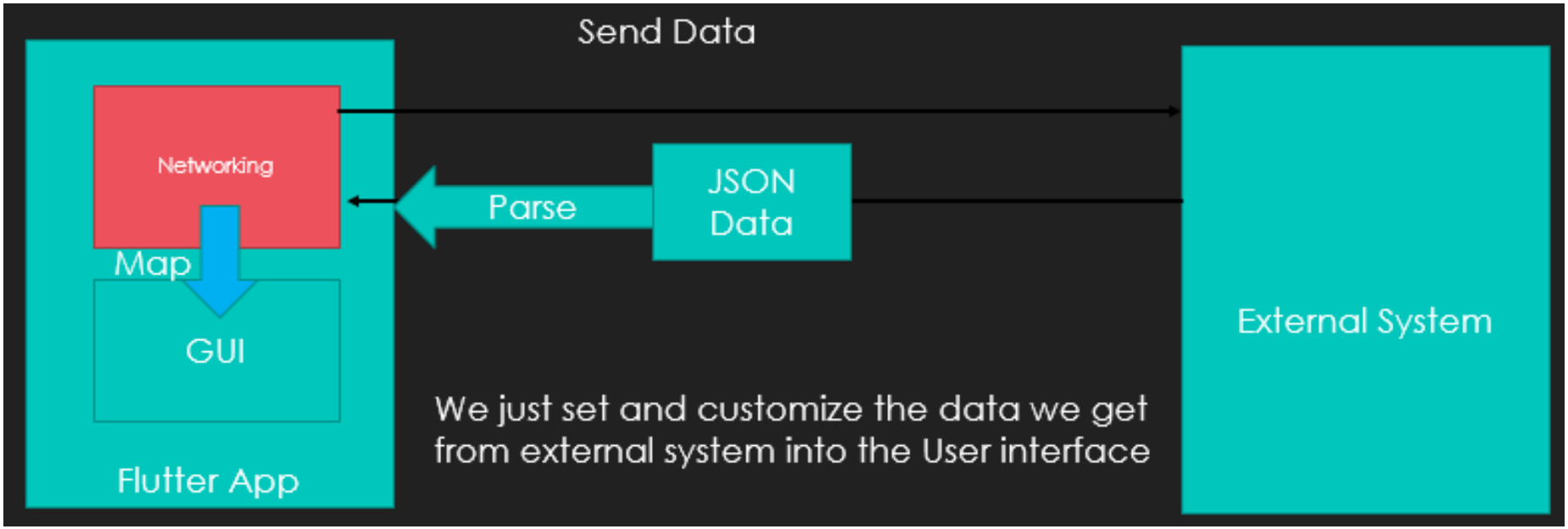
```
Future<void> pengambilData() async {
   var data;
   data=await http.get(
      Uri.parse('https://jsonplaceholder.typicode.com/posts'));

   var dataJson=jsonDecode(data.body);
   for(int i=0;i<dataJson.length;i++) {
      print(dataJson[i]['userID']);
      print(dataJson[i]['id']);
      print(dataJson[i]['title']);
      print(dataJson[i]['body']);
   }

}
```

Read all data

```
Performing hot restart...
Syncing files to device Android SDK built for arm64...
Restarted application in 478ms.
I/flutter ( 6250): null
I/flutter ( 6250): 1
I/flutter ( 6250): sunt aut facere repellat provident occaecati excepturi optio reprehenderit
I/flutter ( 6250): quia et suscipit
I/flutter ( 6250): suscipit recusandae consequuntur expedita et cum
I/flutter ( 6250): reprehenderit molestiae ut ut quas totam
```

Integration with the UI

- Don’t forget to use dialog / snackbar for the result message



Send Data to External System

Send Data to External System (GET)

```
Future<void> pengambilData() async {
    var data;
    data=await http.get(Uri.parse('http://192.168.0.13?Data1=Nilai1&Data2=Nilai2'));
    print(data.body);
}
```

Send :
Data1 with value 'Nilai1'
Data2 with value 'Nilai2'

Sending Data with GET is done by adding the data in the URL :
https://linkurl?namaVariable1=Nilai1&namaVariable2=Nilai2....

Send Data to External System (POST)

```
Future<void> pengambilData() async {
    var data;
    data=await http.post(Uri.parse('http://192.168.0.13', some_data));
    print(data.body);
}
```

Sending Data with POST is done by changing the method to http.post and adding data you want to send in the second parameter.
data = await http.post('alamat eksternal', some_data })

Send Data to External System (POST)

```
Future<void> pengambilData() async {
  var data;
  data=await http.post(Uri.parse('http://192.168.0.13'), headers: <String, String>{
      'Content-Type': 'application/json; charset=UTF-8',
    },
    body: jsonEncode(<String, String>{
      Data1': Nilai1,
    }),
  );

  print(data.body);
}
```

Send :
Data1 with value 'Nilai1' in JSON format

Data is sent in JSON format.
For other format we can modify it in body value.

Any question?

References

- https://flutter.dev
- https://docs.flutter.dev/cookbook/networking/fetch-data
- https://docs.flutter.dev/development/data-and-backend/networking

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THANK YOU